QUALITY ASSURANCE BASELINE

RSCI FYP Portal Test Plan

Final Year Project Management Portal
Complete functional, integration, security, performance, accessibility, recovery, and user-acceptance verification plan

Version	1.0
Date	14 October 2026
Status	Baseline for implementation and acceptance
Companion documents	SRS v1.0 · SDD v1.0 · Database Design v1.0 · API Specification v1.0

Riphah School of Computing & Innovation

DOCUMENT CONTROL

Approval and change control

Field	Value
Document	RSCI FYP Portal Test Plan
Version / date	1.0 / 14 October 2026
Owner	FYP Portal Product and Quality Team
Classification	Internal project documentation
Test basis	SRS v1.0, SDD v1.0, Database Design v1.0, API Specification v1.0, and FYP Framework v1.0

Revision history

Version	Date	Description	Approval
1.0	14 Oct 2026	Initial complete baseline covering M1-M15 and 90 functional test cases.	Pending formal sign-off

> **Policy note.** Values such as group minimum/maximum, supervision cap, lockout threshold, reminder windows, grade publication rules, and retention periods are configurable. Test examples use explicit fixtures; they do not replace institutional approval.

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Page references apply to this baseline and may move when formally approved policy text is inserted.

SECTION 1

Introduction and scope 01

> **Objective.** Prove that the portal correctly controls the complete multi-domain FYP lifecycle, protects academic records, and produces auditable evidence suitable for operational release.

1.1 In scope

- All 15 SRS modules M1-M15, their APIs, database rules, UI workflows, scheduled jobs, reports, exports, and audit events.
- Role-based access control for Super Admin, FYP Coordinator, HOD/management, Supervisor, Co-Supervisor, Student, Internal Evaluator, External Evaluator, and Industry Partner where enabled.
- Cross-module paths from session setup through group formation, proposal, supervision, meetings, deliverables, AI declarations, evaluations, results, and archival.
- Security, performance, accessibility, resilience, backup/restore, compatibility, and privacy checks.

1.2 Out of scope

- Internal accuracy of any third-party plagiarism-detection engine; only request, response, timeout, authorization, and evidence integration are tested.
- Production-data migration execution; migration tooling is tested against anonymized representative copies under a separately approved migration plan.
- Provider infrastructure beyond contractual interfaces, and institutional policy approval itself.

1.3 References

Companion baselines: RSCI FYP Portal SRS v1.0, SDD v1.0, Database Design v1.0, API Specification v1.0; RSCI FYP Framework v1.0.

SECTION 2

Test basis and quality objectives 02

2.1 Requirement identity

This plan uses `FR-Mx-yy` to identify the numbered functional requirements in modules M1-M15. If the formally approved SRS renumbers an item, the traceability table must be reconciled before execution; test intent and evidence remain unchanged.

2.2 Quality objectives

100%
Critical requirement coverage
Every Critical requirement has positive, negative, authorization, and audit evidence where applicable.

0
Open release blockers
No unresolved Critical or High defect at production approval.

500
Concurrent users
Target workload with p95 interactive page/API response below 2 seconds for defined transactions.

AA
Accessibility target
WCAG 2.1 AA spot checks on primary journeys plus automated scanning.

2.3 Assumptions requiring confirmation

- Institutional values are stored as effective-dated configuration, not hard-coded.
- All files pass malware scanning before release from quarantine.
- Time-based rules use a declared institutional timezone.
- Evaluation publication and moderation authorities are explicitly configured.

SECTION 3

Test strategy and levels

03

Level	Purpose and ownership	Minimum evidence
Unit	Developers verify domain rules, validators, policy evaluators, state machines, calculations, and authorization guards in isolation.	Repeatable suite, branch coverage report, mutation or boundary evidence for critical rules.
Integration	Developers and QA verify database constraints, queues, object storage, notifications, exports, and third-party adapters.	Containerized run, rollback checks, idempotency proof, adapter stubs.
API / contract	QA validates OpenAPI schemas, authentication, error envelopes, pagination, idempotency, and consumer/provider compatibility.	Schema report, negative requests, version-compatibility report.
System / E2E	QA executes complete role-specific workflows in a production-like environment.	Case result, screenshots or response captures, database/audit correlation.
UAT	FYP coordinators, supervisors, students, and evaluators validate usability and policy fit.	Signed scenario sheet, deviations, acceptance decision.

3.1 Risk-based emphasis

Highest priority goes to authorization, proposal state transitions, workload caps, immutable submissions, AI-policy flags, grade calculations/publication, audit evidence, and recovery. Medium-risk presentation and convenience behavior follows after academic integrity controls pass.

SECTION 3 CONTINUED

Approach by module and regression

Modules	Primary techniques	Required cross-check
M1-M3	Boundary values, state transition, auth/session tests, CSV fuzzing	Audit, timezone, effective policy snapshot
M4-M7	Catalogue fixtures, constraint tests, concurrency, matching/load rules	Historical version retention and permission scope
M8-M11	Workflow, calculation, file security, policy decision tables	300-hour and 9-hours/week rollups; declaration-to-version link
M12-M15	Rubric math, publication barrier, event idempotency, export reconciliation, tamper tests	Before/after evidence and data minimization

3.2 Regression layers

- **Commit gate:** unit tests, linting, static analysis, migration dry-run, API schema compatibility.
- **Pull-request gate:** changed-module integration tests plus authorization and state-machine smoke.
- **Daily:** full API suite and critical cross-module E2E path.
- **Release candidate:** all 90 functional cases, non-functional suites, backup/restore drill, and UAT regression of accepted journeys.
- **Hotfix:** defect reproduction, adjacent boundary cases, affected-role authorization, and targeted full-module regression.

3.3 Automation principles

Tests must be deterministic, independently seeded, safe to rerun, and traceable to a requirement. Time, queues, mail, malware scanning, and third-party services use controllable test doubles until production-like verification.

SECTION 4

Test environments

04

Environment	Purpose	Required characteristics
Developer	Unit and component feedback	Ephemeral database; local object storage/queue stubs; deterministic clock; synthetic users.
CI integration	Automated migration, API, and integration gates	Fresh isolated stack per run; pinned versions; seeded fixtures; no shared mutable state.
Staging	System, E2E, security, accessibility, and UAT	Production-equivalent topology and configuration shape; masked or synthetic data; outbound notifications sandboxed.
Performance	Load, stress, soak, and capacity	Production-like sizing and observability; isolated from UAT; reversible bulk seed.
Production-like recovery	Backup and restore drill	Encrypted backups, separate restore target, documented recovery point/time measurement.

4.1 Build identity

Every execution records application commit/build, schema migration version, API specification version, feature flags, browser/device matrix, environment configuration hash, seeded dataset version, and test-suite revision.

> **Never test destructively in production.** Grade tampering, audit integrity, malware, injection, stress, and restore exercises use isolated test environments and synthetic data.

SECTION 4 CONTINUED

Test data and evidence

4.2 Minimum seeded dataset

- Two academic sessions: one ACTIVE, one archived; policy versions for group size, supervision cap, target hours, deadlines, and lockout.
- At least two users per role, plus multi-role, disabled, locked, and conflict-of-interest fixtures.
- At least two groups for each of the ten framework project types, including boundary-size groups and unassigned/over-cap scenarios.
- Proposal instances in every workflow state; deliverables on time, late, resubmitted, accepted, and rejected.
- Meetings with mixed attendance; weekly effort totals below, at, and above nine hours; cumulative totals near 300.
- Rubrics with multiple variants and panel sizes; draft, moderated, and published results.

4.3 Privacy and cleanup

Use synthetic names and documents. If production-derived structures are essential, approval, minimization, irreversible masking, access control, and expiry are mandatory. Cleanup jobs remove test files, notifications, tokens, and personal data after the agreed retention period.

4.4 Evidence bundle

For each run retain: plan version, case result, actor/role, input data reference, timestamp/timezone, expected and actual result, screenshots or API evidence, correlated audit/event ID, defect link, and approver. Secrets and raw authentication tokens must never enter evidence.

SECTION 5

Entry and exit criteria

05

Level	Entry	Exit
Unit	Reviewed code; acceptance rules defined	All tests pass; Critical rules have boundary/negative cases; no new Critical static-analysis issue
Integration	Unit exit met; migrations and dependencies available	Interfaces, rollback, idempotency, constraints pass; no open Critical/High integration defect
API	Versioned API specification and auth model approved	Schema, error, permission, pagination and compatibility suites pass
System	Release candidate deployed; stable seed; observability enabled	All Critical and High cases pass; overall pass ≥95%; no blocker; accepted workarounds documented
UAT	System exit met; users trained; scenarios and support ready	All acceptance criteria signed; no unresolved Critical/High policy mismatch
Release	Security, performance, recovery and UAT approval	Go/no-go signed; rollback and monitoring rehearsed; evidence archived

The 95% overall threshold is a proposed release gate and requires project governance approval. It never overrides the zero Critical/High blocker rule.

SECTION 5 CONTINUED

Defect governance and responsibilities

Severity	Definition	Portal example
Critical	Security/academic integrity breach, data loss, unavailable core journey, no safe workaround	Student changes grade; audit can be deleted; result leaks before publication
High	Major function fails or gives materially wrong decision; workaround unacceptable	Seventh supervision allocation accepted; effort total incorrect
Medium	Partial failure with viable workaround; limited scope	One report filter omitted while source data remains correct
Low	Cosmetic or minor usability issue with no integrity impact	Non-blocking alignment or wording defect

5.1 Lifecycle

New → Triaged → Assigned → In Progress → Fixed → Ready for Retest → Verified → Closed. Reopen when reproduction persists. Deferred and Rejected states require reason, owner, risk acceptance, and target release.

5.2 RACI summary

Role	Accountability
QA Lead	Plan, environments, execution control, evidence, reporting, exit recommendation
Developers	Unit/integration suites, defect fixes, diagnostics, test hooks, migration safety
Security/Operations	Threat tests, monitoring, backup/restore, capacity and release readiness
FYP Coordinator	Policy fixtures, UAT leadership, acceptance decision
Supervisors / Students / Evaluators	Role-specific UAT and usability feedback

SECTION 6

Requirement traceability matrix 06

The matrix links each module and its representative numbered functional requirements to executable acceptance cases. Detailed SRS-to-case reconciliation is a release-entry condition.

Module	SRS capability	Requirement IDs	Test cases
M1	User & Identity Management	FR-M1-01..06	TC-UM-01, TC-UM-02, TC-UM-03, TC-UM-04, TC-UM-05, TC-UM-06
M2	Roles, Permissions & Access Control	FR-M2-01..06	TC-RB-01, TC-RB-02, TC-RB-03, TC-RB-04, TC-RB-05, TC-RB-06
M3	Academic Session & Configuration	FR-M3-01..06	TC-SE-01, TC-SE-02, TC-SE-03, TC-SE-04, TC-SE-05, TC-SE-06
M4	Project-Type Catalogue	FR-M4-01..06	TC-PT-01, TC-PT-02, TC-PT-03, TC-PT-04, TC-PT-05, TC-PT-06
M5	Groups & Membership	FR-M5-01..06	TC-GR-01, TC-GR-02, TC-GR-03, TC-GR-04, TC-GR-05, TC-GR-06
M6	Proposal & Approval Workflow	FR-M6-01..06	TC-PR-01, TC-PR-02, TC-PR-03, TC-PR-04, TC-PR-05, TC-PR-06
M7	Supervisor Allocation	FR-M7-01..06	TC-SA-01, TC-SA-02, TC-SA-03, TC-SA-04, TC-SA-05, TC-SA-06
M8	Meetings, Attendance & Logbook	FR-M8-01..06	TC-MT-01, TC-MT-02, TC-MT-03, TC-MT-04, TC-MT-05, TC-MT-06

Coverage rule. A requirement is covered only when its positive behavior, invalid input, role boundary, persistence/audit effect, and relevant concurrency path have evidence. Listing an ID alone is not proof of coverage.

SECTION 6

Requirement traceability matrix 06

The matrix links each module and its representative numbered functional requirements to executable acceptance cases. Detailed SRS-to-case reconciliation is a release-entry condition.

Module	SRS capability	Requirement IDs	Test cases
M9	Milestones & Deliverables	FR-M9-01..06	TC-MS-01, TC-MS-02, TC-MS-03, TC-MS-04, TC-MS-05, TC-MS-06
M10	Document Repository	FR-M10-01..06	TC-DR-01, TC-DR-02, TC-DR-03, TC-DR-04, TC-DR-05, TC-DR-06
M11	AI-Usage Declaration	FR-M11-01..06	TC-AI-01, TC-AI-02, TC-AI-03, TC-AI-04, TC-AI-05, TC-AI-06
M12	Evaluation, Rubrics & Results	FR-M12-01..06	TC-EV-01, TC-EV-02, TC-EV-03, TC-EV-04, TC-EV-05, TC-EV-06
M13	Notifications & Announcements	FR-M13-01..06	TC-NT-01, TC-NT-02, TC-NT-03, TC-NT-04, TC-NT-05, TC-NT-06
M14	Dashboards, Reports & Analytics	FR-M14-01..06	TC-RP-01, TC-RP-02, TC-RP-03, TC-RP-04, TC-RP-05, TC-RP-06
M15	Audit, Compliance & Governance	FR-M15-01..06	TC-AU-01, TC-AU-02, TC-AU-03, TC-AU-04, TC-AU-05, TC-AU-06

Coverage rule. A requirement is covered only when its positive behavior, invalid input, role boundary, persistence/audit effect, and relevant concurrency path have evidence. Listing an ID alone is not proof of coverage.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M1 · User & Identity Management

TC-UM-01 · Authenticate an active user with valid credentials

Critical

M1 User & Identity Management · FR-M1-01

PRECONDITIONS
An active coordinator account exists; no lock is present.

STEPS
1. Open the sign-in page.
2. Enter the registered email and correct password.
3. Submit the form.

EXPECTED RESULT
One authenticated session is created; the coordinator dashboard opens; no credential is exposed in the URL or response body.

TC-UM-02 · Reject invalid credentials without account disclosure

Critical

M1 User & Identity Management · FR-M1-02

PRECONDITIONS
One active account exists for the supplied email.

STEPS
1. Enter the valid email with an incorrect password.
2. Submit, then repeat with an unregistered email.
3. Compare both responses.

EXPECTED RESULT
Both attempts fail with the same generic message and status; no session is issued; each attempt is auditable.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M1 · User & Identity Management

TC-UM-03 · Apply configurable lockout after repeated failures

Critical

M1 User & Identity Management · FR-M1-03

PRECONDITIONS

Lockout threshold is configured to five attempts in the test environment.

STEPS

1. Submit five incorrect passwords for one account.
2. Attempt the correct password before the lock interval expires.
3. Attempt again after an administrator unlocks the account.

EXPECTED RESULT

The account is locked at the configured threshold; pre-unlock login is denied; post-unlock valid login succeeds; lock and unlock are logged.

TC-UM-04 · Enforce password policy and secure reset

High

M1 User & Identity Management · FR-M1-04

PRECONDITIONS

Password policy and a valid single-use reset token are configured.

STEPS

1. Submit passwords that fail length, complexity, and reuse rules.
2. Submit a compliant password.
3. Reuse the consumed reset token.

EXPECTED RESULT

Weak or reused passwords are rejected with field-level guidance; the compliant password is saved securely; the used token cannot be replayed.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M1 · User & Identity Management

TC-UM-05 · Bulk-import valid users from CSV

High

M1 User & Identity Management · FR-M1-05

PRECONDITIONS

Coordinator has import permission; a UTF-8 CSV contains unique required fields for students and supervisors.

STEPS

1. Upload the valid CSV.
2. Review the validation preview.
3. Confirm import and search for two imported users.

EXPECTED RESULT

Preview counts and roles match the file; one atomic import creates all valid users; temporary onboarding state is applied; import summary is downloadable.

TC-UM-06 · Reject malformed or duplicate CSV rows safely

High

M1 User & Identity Management · FR-M1-06

PRECONDITIONS

A CSV includes missing identifiers, an invalid role, and duplicate email rows.

STEPS

1. Upload the malformed CSV.
2. Inspect row-level errors.
3. Cancel the import and search for affected users.

EXPECTED RESULT

Every invalid row is identified by row and field; no partial records are created after cancellation; the original file is retained only per configured retention policy.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M2 · Roles, Permissions & Access Control

TC-RB-01 · Deny a student access to coordinator-only administration

Critical

M2 Roles, Permissions & Access Control · FR-M2-01

PRECONDITIONS

A student session is active; the user-management endpoint is known.

STEPS

1. Call the user-list endpoint with the student session.
2. Attempt to open the administration route directly.
3. Check the audit trail.

EXPECTED RESULT

API and UI requests are denied with 403 or equivalent; no data is returned; the denial is logged without sensitive payloads.

TC-RB-02 · Enforce supervisor scope to assigned groups

Critical

M2 Roles, Permissions & Access Control · FR-M2-02

PRECONDITIONS

Supervisor A is assigned to Group A but not Group B.

STEPS

1. Open Group A records.
2. Change the URL or identifier to Group B.
3. Attempt to download Group B deliverable.

EXPECTED RESULT

Group A is available; both Group B attempts are denied server-side; identifiers do not leak metadata about Group B.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M2 · Roles, Permissions & Access Control

TC-RB-03 · Enforce evaluator access only during assigned evaluation

Critical

M2 Roles, Permissions & Access Control · FR-M2-03

PRECONDITIONS
Evaluator is assigned to one panel and not assigned to another.

STEPS
1. Open the assigned evaluation.
2. Request the unassigned evaluation by ID.
3. Attempt to submit a score for the unassigned project.

EXPECTED RESULT
Assigned work is accessible within the active window; unassigned read and write operations are denied; no score record is created.

TC-RB-04 · Keep role scope isolated per session

Critical

M2 Roles, Permissions & Access Control · FR-M2-04

PRECONDITIONS
One person has both supervisor and evaluator roles.

STEPS
1. Start a supervisor-scoped session.
2. Attempt an evaluator-only action without switching scope.
3. Switch through the approved role selector and retry.

EXPECTED RESULT
The first action is denied; explicit role switching creates a correctly scoped session; permissions and navigation match only the selected role.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M2 · Roles, Permissions & Access Control

TC-RB-05 · Prevent privilege escalation through request tampering

Critical

M2 Roles, Permissions & Access Control · FR-M2-05

PRECONDITIONS
A supervisor session and a captured profile-update request are available.

STEPS
1. Modify the request role field to FYP Coordinator.
2. Submit the altered request.
3. Refresh the profile and permission claims.

EXPECTED RESULT
The server ignores or rejects unauthorized role fields; no role grant occurs; the tampering attempt is recorded.

TC-RB-06 · Apply role revocation to active access

High

M2 Roles, Permissions & Access Control · FR-M2-06

PRECONDITIONS
A user has an active evaluator session.

STEPS
1. Administrator revokes the evaluator role.
2. Use the existing session to read an evaluation.
3. Re-authenticate and inspect available roles.

EXPECTED RESULT
Revoked access stops according to the configured immediate-invalidation policy; fresh authentication excludes the role; revocation is auditable.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M3 · Academic Session & Configuration

TC-SE-01 · Create a valid academic session

High

M3 Academic Session & Configuration · FR-M3-01

PRECONDITIONS Coordinator has configuration permission; no overlapping identifier exists.

STEPS

1. Enter unique name, start date, end date, and status DRAFT.
2. Save the session.
3. Reload its details.

EXPECTED RESULT The session is stored with an immutable identifier; dates and status match input; creation is audited.

TC-SE-02 · Reject an end date not later than the start date

High

M3 Academic Session & Configuration · FR-M3-02

PRECONDITIONS Coordinator is on the session form.

STEPS

1. Set end date equal to start date and submit.
2. Set end date earlier than start date and submit.

EXPECTED RESULT Both submissions are rejected with field-specific validation; no session is created.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M3 · Academic Session & Configuration

TC-SE-03 · Permit only one active session

Critical

M3 Academic Session & Configuration · FR-M3-03

PRECONDITIONS

Session A is ACTIVE and Session B is DRAFT.

STEPS

1. Attempt to activate Session B without closing A.
2. Use the controlled rollover action.
3. Reload both records.

EXPECTED RESULT

Direct activation is blocked; controlled rollover closes or archives A per configuration and activates B atomically; exactly one session is ACTIVE.

TC-SE-04 · Freeze policy snapshot after session activation

High

M3 Academic Session & Configuration · FR-M3-04

PRECONDITIONS

A DRAFT session has group size and supervision-cap settings.

STEPS

1. Activate the session.
2. Edit a governed setting directly.
3. Create an authorized versioned policy change if enabled.

EXPECTED RESULT

Unversioned mutation is denied; any permitted change creates an effective-dated version and audit record without altering historical decisions.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M3 · Academic Session & Configuration

TC-SE-05 · Validate milestone dates against session bounds

High

M3 Academic Session & Configuration · FR-M3-05

PRECONDITIONS

An ACTIVE session has known start and end dates.

STEPS

1. Create a milestone before the session start.
2. Create one after the session end.
3. Create one inside the bounds.

EXPECTED RESULT

Out-of-bound dates are rejected; the in-bound milestone is saved in the session timezone.

TC-SE-06 · Archive a session without deleting history

High

M3 Academic Session & Configuration · FR-M3-06

PRECONDITIONS

A completed session contains groups, evaluations, and documents.

STEPS

1. Archive the session.
2. Search current active records.
3. Open historical reports.

EXPECTED RESULT

Archived records disappear from current-entry workflows but remain read-only and reportable; no linked history is deleted.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M4 · Project-Type Catalogue

TC-PT-01 · Seed all ten framework project types

Critical

M4 Project-Type Catalogue · FR-M4-01

PRECONDITIONS

Fresh database migration and catalogue seeder are available.

STEPS

1. Run migration and seed.
2. List active project types.
3. Compare names, programs, deliverables, and AI policy values with the approved framework.

EXPECTED RESULT

Exactly ten unique types are present: Product Development, Entrepreneurial, Continuation, Research, IoT & Embedded Systems, Case Study, and four Consulting variants; stored policy values match the approved configuration.

TC-PT-02 · Create a project type with complete metadata

High

M4 Project-Type Catalogue · FR-M4-02

PRECONDITIONS

Coordinator has catalogue permission.

STEPS

1. Enter unique code, name, eligible programs, outcomes, deliverables, and AI policy.
2. Save and reopen the type.

EXPECTED RESULT

All fields persist; the type is selectable only for eligible programs and is included in audit history.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M4 · Project-Type Catalogue

TC-PT-03 · Reject duplicate code and missing AI policy

Critical

M4 Project-Type Catalogue · FR-M4-03

PRECONDITIONS

One catalogue type already uses code RESEARCH.

STEPS

1. Create another type with code RESEARCH.
2. Create a new code without ai_policy.

EXPECTED RESULT

Both requests fail; uniqueness and mandatory-policy errors are explicit; no partial record is stored.

TC-PT-04 · Version a catalogue type already in use

High

M4 Project-Type Catalogue · FR-M4-04

PRECONDITIONS

An approved project references the Research type.

STEPS

1. Edit its deliverable template or AI policy.
2. Publish the change for a future session.
3. Open the historical project.

EXPECTED RESULT

A new effective version is created; future sessions use it; the historical project retains its original snapshot.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M4 · Project-Type Catalogue

TC-PT-05 · Deactivate a type without breaking old projects

High

M4 Project-Type Catalogue · FR-M4-05

PRECONDITIONS
A type is referenced by prior-session projects.

STEPS
1. Deactivate the type.
2. Start a new proposal and inspect options.
3. Open an older project of that type.

EXPECTED RESULT
The type is unavailable for new selection but historical records, reports, and policy evidence remain readable.

TC-PT-06 · Restrict catalogue maintenance to authorized roles

Critical

M4 Project-Type Catalogue · FR-M4-06

PRECONDITIONS
Student, supervisor, and coordinator sessions exist.

STEPS
1. Attempt create, edit, and deactivate calls with each session.

EXPECTED RESULT
Student and supervisor attempts are denied server-side; coordinator actions follow configured permission; all denied attempts are logged.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M5 · Groups & Membership

TC-GR-01 · Create a group within configurable size limits

High

M5 Groups & Membership · FR-M5-01

PRECONDITIONS

Active session limits are minimum 2 and maximum 4 for this test.

STEPS

1. Leader creates a group with two eligible students.
2. Add a third and fourth eligible student.
3. Open membership history.

EXPECTED RESULT

Group remains valid at sizes two through four; membership and timestamps are recorded once each.

TC-GR-02 · Reject a fifth group member when maximum is four

Critical

M5 Groups & Membership · FR-M5-02

PRECONDITIONS

A group has four active members and maximum is configured as four.

STEPS

1. Invite a fifth student.
2. Have the student accept, or call the join endpoint directly.

EXPECTED RESULT

The operation is rejected atomically; group remains at four; the reason references the configured maximum.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M5 · Groups & Membership

TC-GR-03 · Reject solo submission when minimum is two

Critical

M5 Groups & Membership · FR-M5-03

PRECONDITIONS
A one-member draft group exists; minimum is configured as two.

STEPS
1. Attempt to submit the group for approval.

EXPECTED RESULT
Submission is blocked; the draft remains editable; validation states that one additional eligible member is required.

TC-GR-04 · Prevent duplicate membership across active groups

Critical

M5 Groups & Membership · FR-M5-04

PRECONDITIONS
Student X belongs to active Group A.

STEPS
1. Invite Student X to Group B.
2. Attempt direct join to Group B.

EXPECTED RESULT
Both paths reject the duplicate active membership under the same session; no second membership row is created.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M5 · Groups & Membership

TC-GR-05 · Transfer group leadership with consent and history

High

M5 Groups & Membership · FR-M5-05

PRECONDITIONS

Group has Leader A and Member B.

STEPS

1. Leader A initiates transfer to Member B.
2. Member B accepts.
3. Attempt a leader-only action as both users.

EXPECTED RESULT

B becomes sole leader; A becomes member; permissions change immediately; transfer initiator, accepter, and timestamps are retained.

TC-GR-06 · Guarantee unique non-guessable group codes

High

M5 Groups & Membership · FR-M5-06

PRECONDITIONS

Concurrent group creation is available.

STEPS

1. Create 100 groups concurrently in the test database.
2. Check code uniqueness and allowed format.
3. Attempt to submit a duplicate code through the API.

EXPECTED RESULT

All generated codes are unique; duplicate insertion is rejected by database and service constraints; error handling does not expose internals.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M6 · Proposal & Approval Workflow

TC-PR-01 · Complete the valid proposal approval path

Critical

M6 Proposal & Approval Workflow · FR-M6-01

PRECONDITIONS	Eligible group has complete proposal data and selected type.
STEPS	1. Save DRAFT. 2. Submit to reach SUBMITTED. 3. Supervisor accepts review to reach UNDER_SUPERVISOR_REVIEW. 4. Forward to coordinator, then approve.
EXPECTED RESULT	States occur only in order: DRAFT, SUBMITTED, UNDER_SUPERVISOR_REVIEW, UNDER_COORDINATOR_REVIEW, APPROVED; each transition records actor, time, and comment.

TC-PR-02 · Reject illegal proposal state transition with WF_001

Critical

M6 Proposal & Approval Workflow · FR-M6-02

PRECONDITIONS	Proposal is in DRAFT.
STEPS	1. Call the transition endpoint requesting APPROVED. 2. Reload state and history.
EXPECTED RESULT	Request fails with domain error WF_001; state remains DRAFT; illegal attempt is auditable.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M6 · Proposal & Approval Workflow

TC-PR-03 · Run supervisor resubmission-required loop

High

M6 Proposal & Approval Workflow · FR-M6-03

PRECONDITIONS

Proposal is UNDER_SUPERVISOR_REVIEW.

STEPS

1. Supervisor selects RESUBMISSION_REQUIRED with actionable comments.
2. Student edits and resubmits.
3. Supervisor reopens the new version.

EXPECTED RESULT

A new editable revision is created; previous content is immutable; resubmission returns to the configured review state with linked comments.

TC-PR-04 · Run coordinator rejection path

High

M6 Proposal & Approval Workflow · FR-M6-04

PRECONDITIONS

Proposal is UNDER_COORDINATOR_REVIEW.

STEPS

1. Coordinator rejects with mandatory reason.
2. Student attempts to edit.
3. Coordinator reopens only if policy permits.

EXPECTED RESULT

State becomes REJECTED; student editing follows configured terminal-state policy; any reopening is permission-checked and logged.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M6 · Proposal & Approval Workflow

TC-PR-05 · Preserve proposal versions and checksums

High

M6 Proposal & Approval Workflow · FR-M6-05

PRECONDITIONS	Proposal version 1 is SUBMITTED.
STEPS	1. Request changes, upload modified proposal as version 2, and resubmit. 2. Download both versions and inspect metadata.
EXPECTED RESULT	Both versions remain available to authorized users; sequence, uploader, timestamp, and checksum are distinct and correct.

TC-PR-06 · Resolve concurrent review transitions safely

Critical

M6 Proposal & Approval Workflow · FR-M6-06

PRECONDITIONS	Two authorized reviewers load the same review state.
STEPS	1. Reviewer A forwards the proposal. 2. Reviewer B submits a stale resubmission-required action.
EXPECTED RESULT	Only one transition commits; stale request receives a conflict response; history contains no fork or duplicate state.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M7 · Supervisor Allocation

TC-SA-01 · Allocate supervisor within configured load cap

High

M7 Supervisor Allocation · FR-M7-01

PRECONDITIONS	Maximum active load is configured as six; supervisor has five active groups.
STEPS	1. Allocate a sixth eligible group. 2. Refresh workload dashboard.
EXPECTED RESULT	Allocation succeeds; active load becomes six; group and supervisor views agree.

TC-SA-02 · Reject seventh allocation when maximum is six

Critical

M7 Supervisor Allocation · FR-M7-02

PRECONDITIONS	Supervisor already has six active groups and cap is six.
STEPS	1. Attempt a seventh allocation through UI. 2. Repeat through direct API call.
EXPECTED RESULT	Both attempts fail; load remains six; coordinator sees the capacity reason.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M7 · Supervisor Allocation

TC-SA-03 · Filter supervisor suggestions by expertise

High

M7 Supervisor Allocation · FR-M7-03

PRECONDITIONS
Groups and supervisors have normalized expertise tags.

STEPS
1. Open allocation suggestions for an AI/Data Analytics project.
2. Inspect ranking and excluded profiles.

EXPECTED RESULT
Eligible matching supervisors rank above weaker matches; filters are explainable; no ineligible supervisor is auto-assigned.

TC-SA-04 · Reassign supervisor while retaining history

Critical

M7 Supervisor Allocation · FR-M7-04

PRECONDITIONS
Group is assigned to Supervisor A with meeting history.

STEPS
1. Coordinator reassigns to Supervisor B with reason.
2. Open current assignment and assignment history.

EXPECTED RESULT
B becomes current owner; A remains in historical record; past meetings retain original actor attribution; reason is logged.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M7 · Supervisor Allocation

TC-SA-05 · Prevent conflict-of-interest allocation

Critical

M7 Supervisor Allocation · FR-M7-05

PRECONDITIONS

Evaluator or supervisor has a declared conflict with one group.

STEPS

1. Attempt manual allocation.
2. Attempt bulk auto-allocation.

EXPECTED RESULT

Both paths exclude or block the conflicted assignment; override, if institutionally enabled, requires elevated permission and documented reason.

TC-SA-06 · Make bulk allocation atomic and explain failures

High

M7 Supervisor Allocation · FR-M7-06

PRECONDITIONS

Batch contains valid rows, one over-cap supervisor, and one unknown group.

STEPS

1. Validate the batch.
2. Confirm without correcting errors.
3. Correct errors and confirm again.

EXPECTED RESULT

Validation identifies each row; invalid confirmation creates no partial allocations; corrected batch commits once with summary.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M8 · Meetings, Attendance & Logbook

TC-MT-01 · Schedule meeting with valid participants and time

High

M8 Meetings, Attendance & Logbook · FR-M8-01

PRECONDITIONS Approved group and assigned supervisor exist.

STEPS

1. Create meeting with agenda, start, duration, and participants.
2. Save and open calendar views.

EXPECTED RESULT One meeting is created in the session timezone; participants and agenda match; notifications are queued once.

TC-MT-02 · Mark attendance for each participant

High

M8 Meetings, Attendance & Logbook · FR-M8-02

PRECONDITIONS Scheduled meeting has occurred.

STEPS

1. Supervisor marks present, absent, and excused statuses.
2. Save and reload attendance.

EXPECTED RESULT Exactly one status per participant persists with marker and timestamp; unauthorized students cannot alter attendance.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M8 · Meetings, Attendance & Logbook

TC-MT-03 · Edit minutes before sign-off and lock after sign-off

Critical

M8 Meetings, Attendance & Logbook · FR-M8-03

PRECONDITIONS

Meeting minutes are in DRAFT.

STEPS

1. Student edits minutes.
2. Supervisor signs off.
3. Student and supervisor attempt further edits.

EXPECTED RESULT

Pre-sign-off edit succeeds; sign-off records actor and time; later edits are denied and signed content remains immutable, with an amendment workflow if configured.

TC-MT-04 · Roll up effort hours toward 300-hour target

Critical

M8 Meetings, Attendance & Logbook · FR-M8-04

PRECONDITIONS

Student has approved log entries of 8.0, 9.5, and 7.5 hours.

STEPS

1. Approve all entries.
2. Open student and group effort dashboards.
3. Export the logbook.

EXPECTED RESULT

Approved total equals 25.0 hours in every view and export; duplicates and rejected entries are excluded; progress uses configurable 300-hour target.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M8 · Meetings, Attendance & Logbook

TC-MT-05 · Calculate the 9-hours-per-week indicator correctly

High

M8 Meetings, Attendance & Logbook · FR-M8-05

PRECONDITIONS
Week uses institutional timezone; approved entries total 8.5 hours.

STEPS
1. Open weekly indicator.
2. Add and approve a 0.5-hour entry.
3. Move an entry across the week boundary.

EXPECTED RESULT
Indicator shows below target at 8.5, meets target at 9.0, then recalculates against the correct week after the boundary change.

TC-MT-06 · Export complete, authorized logbook

High

M8 Meetings, Attendance & Logbook · FR-M8-06

PRECONDITIONS
Group has meetings, attendance, signed minutes, and effort entries.

STEPS
1. Supervisor exports PDF and CSV.
2. Student exports own group logbook.
3. Unassigned supervisor requests the export.

EXPECTED RESULT
Authorized exports honor filters and contain matching totals; unassigned request is denied; export generation is logged.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M9 · Milestones & Deliverables

TC-MS-01 · Generate checklist by project type

Critical

M9 Milestones & Deliverables · FR-M9-01

PRECONDITIONS

Templates exist for all active project types.

STEPS

1. Create a Research project and an Entrepreneurial project.
2. Generate milestone checklists.

EXPECTED RESULT

Research includes literature/research proposal/experimental design/paper draft; Entrepreneurial includes MVP/business plan/BMC/market research/financial feasibility/pitch deck; irrelevant templates are absent.

TC-MS-02 · Submit a required deliverable before deadline

High

M9 Milestones & Deliverables · FR-M9-02

PRECONDITIONS

A group has an open required milestone.

STEPS

1. Upload an allowed file and declaration.
2. Submit the deliverable.
3. Open supervisor review queue.

EXPECTED RESULT

Submission records version, checksum, submitter, and on-time status; one review item appears.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M9 · Milestones & Deliverables

TC-MS-03 · Flag a late deliverable consistently

High

M9 Milestones & Deliverables · FR-M9-03

PRECONDITIONS

Deadline has passed in the session timezone.

STEPS

1. Submit an otherwise valid deliverable.
2. Open group view, supervisor queue, and report export.

EXPECTED RESULT

Submission is accepted or blocked according to configured policy; when accepted, late=true and lateness duration agree in all views.

TC-MS-04 · Version a resubmitted deliverable

Critical

M9 Milestones & Deliverables · FR-M9-04

PRECONDITIONS

Version 1 is marked RESUBMISSION_REQUIRED.

STEPS

1. Upload corrected content.
2. Submit version 2.
3. Attempt to overwrite version 1.

EXPECTED RESULT

Version 2 links to version 1 and enters review; version 1 is immutable and downloadable to authorized users.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M9 · Milestones & Deliverables

TC-MS-05 · Enforce review-loop transitions

Critical

M9 Milestones & Deliverables · FR-M9-05

PRECONDITIONS

Deliverable is SUBMITTED.

STEPS

1. Supervisor opens review, requests resubmission, receives version 2, and accepts it.
2. Attempt direct SUBMITTED to ACCEPTED as student.

EXPECTED RESULT

Reviewer path follows configured states with comments; student transition is denied with WF_001 or equivalent; complete history remains.

TC-MS-06 · Handle deadline extension without corrupting evidence

High

M9 Milestones & Deliverables · FR-M9-06

PRECONDITIONS

One group has an approved extension.

STEPS

1. Record new deadline with reason and approver.
2. Submit between original and extended deadlines.
3. Compare reports.

EXPECTED RESULT

Submission is not late for the approved group; original deadline, new deadline, reason, and approver remain visible; other groups are unaffected.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M10 · Document Repository

TC-DR-01 · Upload an allowed document securely

High

M10 Document Repository · FR-M10-01

PRECONDITIONS

User has write access; PDF limit is configured.

STEPS

1. Upload a valid PDF within size limit.
2. Wait for malware scan.
3. Download the approved file.

EXPECTED RESULT

File is quarantined until scan passes, then stored outside executable paths; metadata and checksum match; download uses authorization checks.

TC-DR-02 · Reject executable and disguised file uploads

Critical

M10 Document Repository · FR-M10-02

PRECONDITIONS

Upload endpoint is available.

STEPS

1. Upload an executable with changed extension.
2. Upload a double-extension file.
3. Upload a file with mismatched MIME signature.

EXPECTED RESULT

All are rejected by content inspection; no executable is served; event is logged and security monitoring is notified where configured.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M10 · Document Repository

TC-DR-03 · Block IDOR access to another group document

Critical

M10 Document Repository · FR-M10-03

PRECONDITIONS Student A can download Group A document; Group B document ID is known.

STEPS

1. Replace document ID with Group B value.
2. Try metadata and download endpoints.

EXPECTED RESULT Both requests are denied without revealing filename, owner, or existence; no signed URL is issued.

TC-DR-04 · Preserve document version and checksum history

High

M10 Document Repository · FR-M10-04

PRECONDITIONS Version 1 exists and user may submit a revision.

STEPS

1. Upload version 2.
2. Compare metadata and checksums.
3. Download version 1.

EXPECTED RESULT Both immutable versions remain linked; current marker points to version 2; audit identifies uploader and timestamp.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M10 · Document Repository

TC-DR-05 · Apply retention and legal-hold rules

High

M10 Document Repository · FR-M10-05

PRECONDITIONS

A test document meets configured retention age; another is under legal hold.

STEPS

1. Run retention job.
2. Attempt deletion of held document.
3. Inspect job report.

EXPECTED RESULT

Eligible document follows configured archive/delete action; held document is preserved; decisions and failures are auditable.

TC-DR-06 · Recover interrupted upload without partial record

Medium

M10 Document Repository · FR-M10-06

PRECONDITIONS

Large upload can be interrupted.

STEPS

1. Interrupt transfer before completion.
2. Restart or resume according to supported protocol.
3. Inspect repository and temporary storage.

EXPECTED RESULT

No incomplete document appears as valid; retry produces one complete record; abandoned chunks are cleaned by policy.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M11 · AI-Usage Declaration

TC-AI-01 · Require AI declaration on every submission

Critical

M11 AI-Usage Declaration · FR-M11-01

PRECONDITIONS
A group has a deliverable ready to submit.

STEPS
1. Omit the AI declaration and submit.
2. Add a complete declaration and resubmit.

EXPECTED RESULT
First request is rejected with field-level error and no submitted version; second succeeds and links declaration to that exact version.

TC-AI-02 · Flag Research project with EXTENSIVE usage

Critical

M11 AI-Usage Declaration · FR-M11-02

PRECONDITIONS
Research project policy disallows extensive assistance in core work/documentation.

STEPS
1. Declare usage level EXTENSIVE with details.
2. Submit and inspect review queue.

EXPECTED RESULT
Submission is retained for review but policy_flag=true with matched policy rule and visible reviewer warning; no automatic misconduct conclusion is recorded.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M11 · AI-Usage Declaration

TC-AI-03 · Do not flag Entrepreneurial EXTENSIVE usage

Critical

M11 AI-Usage Declaration · FR-M11-03

PRECONDITIONS
Entrepreneurial project policy permits extensive AI use.

STEPS
1. Declare EXTENSIVE usage with required tool/purpose details.
2. Submit and inspect policy evaluation.

EXPECTED RESULT
Submission succeeds with policy_flag=false unless another configured rule is violated; declaration remains auditable.

TC-AI-04 · Capture supervisor attestation

High

M11 AI-Usage Declaration · FR-M11-04

PRECONDITIONS
Submitted declaration awaits supervisor attestation.

STEPS
1. Supervisor reviews and attests.
2. Reload declaration and export evidence.

EXPECTED RESULT
Attestation stores supervisor, decision, timestamp, and optional comment; student cannot forge or edit it.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M11 · AI-Usage Declaration

TC-AI-05 · Re-evaluate policy using project snapshot

Critical

M11 AI-Usage Declaration · FR-M11-05

PRECONDITIONS	A historical project has an older AI policy snapshot; catalogue policy later changes.
STEPS	1. Open historical declaration. 2. Run compliance report. 3. Submit a new declaration in a later session.
EXPECTED RESULT	Historical decision uses original snapshot; new session uses current version; report identifies applied policy version.

TC-AI-06 · Protect declaration edits after submission

High

M11 AI-Usage Declaration · FR-M11-06

PRECONDITIONS	Declaration is linked to a submitted version.
STEPS	1. Student attempts direct edit. 2. Use formal amendment flow with reason.
EXPECTED RESULT	Direct edit is denied; amendment creates linked version, preserves original, and triggers re-attestation if required.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M12 · Evaluation, Rubrics & Results

TC-EV-01 · Create rubric whose criterion weights sum to 100

Critical

M12 Evaluation, Rubrics & Results · FR-M12-01

PRECONDITIONS Coordinator can create a rubric variant.

STEPS

1. Add criteria totaling 100 percent.
2. Publish rubric.
3. Create another totaling 95 percent and publish.

EXPECTED RESULT The 100-percent rubric publishes; the 95-percent rubric is rejected with calculated total; draft remains editable.

TC-EV-02 · Reject score above criterion maximum

Critical

M12 Evaluation, Rubrics & Results · FR-M12-02

PRECONDITIONS Evaluator has assigned rubric criterion maximum 10.

STEPS

1. Enter score 11 and save.
2. Enter score 10 and save.

EXPECTED RESULT Score 11 is rejected server-side; score 10 persists; no invalid score enters aggregation.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M12 · Evaluation, Rubrics & Results

TC-EV-03 · Aggregate panel scores using configured method

Critical

M12 Evaluation, Rubrics & Results · FR-M12-03

PRECONDITIONS

Three panel members submit 70, 80, and 90 under equal weighting.

STEPS

1. Finalize all three score sheets.
2. Run aggregation.
3. Open result details.

EXPECTED RESULT

Aggregate is 80.00 with transparent member contributions and rounding; incomplete panel status clears only after required submissions.

TC-EV-04 · Log moderation adjustment with before and after

Critical

M12 Evaluation, Rubrics & Results · FR-M12-04

PRECONDITIONS

Aggregate result is 80 and moderator has permission.

STEPS

1. Change moderated score to 82 with mandatory reason.
2. Open audit and result history.

EXPECTED RESULT

Current moderated score is 82; original 80, new 82, actor, reason, and timestamp are immutable in audit.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M12 · Evaluation, Rubrics & Results

TC-EV-05 · Hide results until PUBLISHED

Critical

M12 Evaluation, Rubrics & Results · FR-M12-05

PRECONDITIONS Result is DRAFT or MODERATED.

STEPS

1. Student opens result page and calls result API.
2. Coordinator publishes.
3. Student retries.

EXPECTED RESULT Before publication no score or ranking is exposed; after publication the authorized student sees only permitted result details.

TC-EV-06 · Select rubric variant by project type and stage

High

M12 Evaluation, Rubrics & Results · FR-M12-06

PRECONDITIONS Research and Entrepreneurial rubric variants are active for Final Demo.

STEPS

1. Create evaluations for each type.
2. Inspect assigned rubric and criteria.

EXPECTED RESULT Each evaluation receives the correct effective variant; Research criteria emphasize research rigor and Entrepreneurial criteria include validated business/MVP outcomes as configured.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M13 · Notifications & Announcements

TC-NT-01 · Notify participants when a meeting is scheduled

High

M13 Notifications & Announcements · FR-M13-01

PRECONDITIONS Notification channels and preferences are configured.

STEPS

1. Schedule a meeting.
2. Inspect in-app queue and permitted outbound channel stub.

EXPECTED RESULT One notification per intended participant contains correct meeting time and link; creator is not duplicated; event correlation ID is retained.

TC-NT-02 · Send deadline reminders without duplication

High

M13 Notifications & Announcements · FR-M13-02

PRECONDITIONS A milestone reminder is configured at 48 hours.

STEPS

1. Advance test clock to reminder time.
2. Run scheduler twice.

EXPECTED RESULT Exactly one reminder per recipient and milestone is recorded; second run is idempotent.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M13 · Notifications & Announcements

TC-NT-03 · Notify evaluators of assignment and changes

High

M13 Notifications & Announcements · FR-M13-03

PRECONDITIONS Evaluation panel is created, then time is updated.

STEPS

1. Assign evaluator.
2. Change evaluation time.
3. Inspect notification history.

EXPECTED RESULT Assignment and update generate distinct, correct notifications; superseded details are clearly identified.

TC-NT-04 · Notify student only after result publication

Critical

M13 Notifications & Announcements · FR-M13-04

PRECONDITIONS Result moves from DRAFT to PUBLISHED.

STEPS

1. Observe events before publication.
2. Publish result.
3. Open notification as another student.

EXPECTED RESULT No premature result alert occurs; owner receives one alert after publication; another student cannot access linked result.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M13 · Notifications & Announcements

TC-NT-05 · Target announcements to intended audience

Critical

M13 Notifications & Announcements · FR-M13-05

PRECONDITIONS	Coordinator can target session, program, role, or specific groups.
STEPS	1. Send announcement to Research groups in one session. 2. Inspect inboxes for included and excluded users.
EXPECTED RESULT	All and only resolved recipients receive it; audience definition and count are stored; no cross-session leakage occurs.

TC-NT-06 · Honor preferences without suppressing mandatory notices

High

M13 Notifications & Announcements · FR-M13-06

PRECONDITIONS	User disables optional email but mandatory academic alerts are configured.
STEPS	1. Trigger optional announcement and mandatory deadline change. 2. Inspect delivery channels.
EXPECTED RESULT	Optional email is suppressed while in-app copy follows policy; mandatory notice uses required channel; preference decisions are recorded.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M14 · Dashboards, Reports & Analytics

TC-RP-01 · Calculate at-risk group using configured rules

Critical

M14 Dashboards, Reports & Analytics · FR-M14-01

PRECONDITIONS Rule flags missed milestone or weekly effort below threshold; fixtures cover both.

STEPS

1. Refresh dashboard.
2. Compare source records with at-risk list.

EXPECTED RESULT Every fixture meeting the rule is flagged once with explainable reason; compliant groups are absent; calculation time is shown.

TC-RP-02 · Calculate evaluation progress percentage

High

M14 Dashboards, Reports & Analytics · FR-M14-02

PRECONDITIONS 20 required evaluations exist; 15 are finalized.

STEPS

1. Open progress KPI.
2. Finalize one more evaluation and refresh.

EXPECTED RESULT KPI is 75 percent initially and 80 percent after finalization; denominator excludes cancelled evaluations according to policy.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M14 · Dashboards, Reports & Analytics

TC-RP-03 · Apply session filters consistently

High

M14 Dashboards, Reports & Analytics · FR-M14-03

PRECONDITIONS Two sessions contain distinct records.

STEPS

1. Filter dashboard to Session A.
2. Export CSV and PDF.
3. Compare visible and exported totals.

EXPECTED RESULT All widgets and exports contain only Session A; filter label is printed in both exports; totals match source query.

TC-RP-04 · Protect sensitive dashboard drill-downs

Critical

M14 Dashboards, Reports & Analytics · FR-M14-04

PRECONDITIONS Coordinator and supervisor sessions exist.

STEPS

1. Coordinator opens institution-wide KPI.
2. Supervisor requests same drill-down.
3. Supervisor opens assigned-group detail.

EXPECTED RESULT Coordinator access follows role; supervisor institution-wide request is denied or scoped; assigned-group detail is permitted.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M14 · Dashboards, Reports & Analytics

TC-RP-05 · Generate accurate CSV export

High

M14 Dashboards, Reports & Analytics · FR-M14-05

PRECONDITIONS Filtered report has known Unicode names, dates, nulls, and totals.

STEPS

1. Export CSV.
2. Open with UTF-8 parser.
3. Recompute totals from rows.

EXPECTED RESULT Headers, encoding, quoting, timezone, null representation, and row count are correct; recomputed totals equal dashboard.

TC-RP-06 · Generate readable PDF report

High

M14 Dashboards, Reports & Analytics · FR-M14-06

PRECONDITIONS A multi-page filtered report is available.

STEPS

1. Export PDF.
2. Inspect every page, headers, footers, filters, and totals.

EXPECTED RESULT PDF is complete, readable, paginated, and contains applied filters and generation time; no row is clipped or omitted.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M15 · Audit, Compliance & Governance

TC-AU-01 · Prevent audit-log update and delete

Critical

M15 Audit, Compliance & Governance · FR-M15-01

PRECONDITIONS

Audit entries exist; administrator session is active.

STEPS

1. Attempt update via API.
2. Attempt delete via API and direct repository method used in tests.
3. Reload entries.

EXPECTED RESULT

No supported path mutates or deletes entries; attempts fail and are themselves logged or alerted; original hashes/content remain.

TC-AU-02 · Capture grade change before and after values

Critical

M15 Audit, Compliance & Governance · FR-M15-02

PRECONDITIONS

Published or moderated grade is 78; authorized change workflow exists.

STEPS

1. Change grade to 82 with reason.
2. Open grade history and audit export.

EXPECTED RESULT

Audit contains protected project/evaluation reference, before=78, after=82, actor, reason, timestamp, and correlation ID.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M15 · Audit, Compliance & Governance

TC-AU-03 · Record authentication and authorization events

High

M15 Audit, Compliance & Governance · FR-M15-03

PRECONDITIONS Success, failure, lockout, and denied-access fixtures are available.

STEPS

1. Trigger each event.
2. Search audit by user, event type, and time.

EXPECTED RESULT Each event is searchable with outcome and safe metadata; passwords, tokens, and document contents are absent.

TC-AU-04 · Verify tamper-evident audit chain

Critical

M15 Audit, Compliance & Governance · FR-M15-04

PRECONDITIONS Test environment supports integrity verification.

STEPS

1. Export a valid verification report.
2. Tamper with a disposable database copy.
3. Run verifier again.

EXPECTED RESULT Valid chain passes; altered copy fails at the affected entry or segment; production data is never modified by the test.

SECTION 7 · FUNCTIONAL ACCEPTANCE SUITE

M15 · Audit, Compliance & Governance

TC-AU-05 · Enforce audit viewing permission and masking

Critical

M15 Audit, Compliance & Governance · FR-M15-05

PRECONDITIONS

Coordinator, auditor/authorized admin, supervisor, and student sessions exist.

STEPS

1. Request audit search with each role.
2. Inspect personal-data fields.

EXPECTED RESULT

Only configured governance roles can search institution-wide logs; other roles are denied or limited; sensitive fields are masked according to policy.

TC-AU-06 · Export audit evidence with stable filters

High

M15 Audit, Compliance & Governance · FR-M15-06

PRECONDITIONS

Authorized user applies actor, module, date, and event filters.

STEPS

1. Export evidence.
2. Re-run export with same fixed dataset.
3. Verify checksum and row count.

EXPECTED RESULT

Export includes filter statement, generation metadata, ordered entries, and checksum; row counts match search; same dataset yields consistent content ordering.

SECTION 8

Performance and capacity 08

Targets below are proposed engineering acceptance targets and must be approved against infrastructure sizing and service objectives.

ID	Workload and method	Acceptance target
NFR-PERF-01	k6/JMeter ramp to 500 concurrent authenticated users across dashboard, proposal, meeting, deliverable, and report-read mix	p95 under 2 s; error rate under 1%; no authorization or data-integrity error
NFR-PERF-02	Submission spike at deadline: 100 concurrent document submissions with malware-scan queue stub	No lost/duplicate submission; accepted request acknowledged within approved limit; queue drains without retry storm
NFR-PERF-03	Large coordinator CSV import and report export at agreed maximum row count	Completes within approved batch SLA; interactive traffic remains responsive; memory bounded
NFR-PERF-04	8-hour soak at 60% expected peak	No material memory/connection growth; stable p95; no duplicate scheduled notifications
NFR-PERF-05	Stress beyond target until controlled degradation	Rate limits/backpressure protect integrity; recovery occurs without restart or data repair

Observability

Capture request rate, latency percentiles, errors, saturation, database wait/slow queries, queue depth, object-store latency, cache hit rate, and business transaction success. Correlate by test-run ID; do not log secrets.

SECTION 8 CONTINUED

Security verification

Area	Executable checks	Pass condition
Authentication	Brute force, session fixation, token replay, logout invalidation, reset-token reuse	No bypass; lockout/rate-limit and invalidation follow configuration
Broken access / IDOR	Change group, document, evaluation, result, meeting, and audit identifiers across roles	Server denies every out-of-scope object without metadata leak
Injection	SQL/NoSQL injection strings in search, import, filters, comments, filenames, and report parameters	Parameterized handling; no query manipulation, stack trace, or stored payload execution
Grade tampering	Client-field changes, stale writes, direct API calls, mass assignment, replay	Only authorized workflow succeeds; before/after audit immutable
File abuse	Executable, macro, polyglot, oversize, zip-bomb, path traversal, MIME mismatch	Quarantine and content inspection prevent execution or unauthorized retrieval
Web attacks	Stored/reflected XSS, CSRF, insecure headers, clickjacking, open redirect	OWASP-recommended defenses verified; no exploitable path
Secrets/privacy	Logs, exports, errors, cache, signed URL expiry, data minimization	No credential leakage; data access and retention follow policy

Use authorized, isolated security testing only. Record evidence and remediation; independent penetration testing is recommended before first production release.

SECTION 8 CONTINUED

Usability, accessibility and compatibility

WCAG 2.1 AA spot-check journeys

- Keyboard-only: sign in, form group, submit proposal, log meeting, upload deliverable, evaluate, publish result, run report.
- Visible focus, logical tab order, skip navigation, semantic headings, labels, instructions, and accessible validation summaries.
- Contrast, 200% zoom/reflow, text spacing, target size, non-color status cues, and reduced-motion behavior.
- Screen-reader names/roles/states for dialogs, tables, date pickers, upload progress, rubric controls, and notifications.
- Document exports have meaningful reading order and table headers where technically supported.

Responsive and browser matrix

Verify current institution-supported desktop browsers and representative Android/iOS mobile widths. Primary student/supervisor tasks must work at 320 CSS pixels without horizontal page scrolling except intrinsically wide data tables, which require accessible alternatives.

Usability acceptance

- Users can always identify current session, group, project type, state, next required action, and deadline.
- Destructive/irreversible actions require clear confirmation and permission.
- Errors preserve safe input, identify the affected field, and explain recovery without exposing internals.

SECTION 8 CONTINUED

Backup, restore and resilience

ID	Drill	Acceptance evidence
NFR-REC-01	Create encrypted backup containing relational data, object metadata, and configuration references; restore to isolated environment.	Restore completes within approved RTO; measured data loss is within approved RPO; checksums and row counts reconcile.
NFR-REC-02	Restore proposal, signed minutes, deliverable versions, declarations, scores, publications, and audit chain.	Cross-record links, permissions, version order, files, and audit verification remain correct.
NFR-RES-01	Interrupt database, queue, object store, and notification provider separately during safe test transactions.	Transactions roll back or retry idempotently; user sees recoverable status; no duplicate grades, submissions, or notices.
NFR-RES-02	Run scheduled job twice and resume after partial execution.	Idempotency keys prevent duplicate reminders, reports, and state changes.

Operational acceptance

Runbooks must identify owner, escalation route, health indicators, alert thresholds, backup status, restore steps, rollback procedure, maintenance mode, and manual reconciliation queries. A backup is not accepted until a restore is successfully demonstrated.

SECTION 9

User acceptance testing 09

9.1 Role-based UAT scenarios

Participant	Acceptance journey
FYP Coordinator	Configure session, verify catalogue/policies, import users, approve proposal, allocate supervisor, publish rubric, monitor progress, export report.
Supervisor	Review proposal, conduct/sign meeting, approve effort, review deliverable and AI declaration, complete continuous assessment.
Student	Form group, select type, submit/resubmit proposal, maintain logbook, upload deliverable and declaration, view published result.
Evaluator	Access assigned project only, score valid rubric, finalize sheet, respond to moderation request.
Management / authorized auditor	Review KPI, drill down within permission, verify grade change and audit evidence.

9.2 Acceptance criteria

- All role journeys complete using approved institutional terminology and policies.
- No unauthorized data or action is observed; calculations and exports reconcile to source records.
- Critical accessibility barriers are absent; help and error messages enable recovery.
- No unresolved Critical/High defect or policy ambiguity affects release.
- Training, runbooks, support, privacy, retention, backup, and rollback arrangements are approved.

SECTION 9 CONTINUED

Release decision and sign-off

9.3 Go / no-go evidence

- Traceability coverage report and signed test summary.
- All 90 cases executed for the release candidate; deviations risk-accepted in writing.
- Security, performance, accessibility, recovery, and UAT reports approved.
- Database migration/rollback rehearsal and API compatibility gate passed.
- Monitoring dashboards, alerts, support roster, backups, and incident runbooks ready.

9.4 Sign-off template

Role	Name	Decision / conditions	Signature	Date
FYP Coordinator / Product Owner				
QA Lead				
Technical Lead				
Security / Operations				
Institutional Approver				

> **Decision rule.** Conditional approval must list every condition, owner, due date, verification method, and rollback trigger.

APPENDIX A

Schedule and reporting cadence

A

Phase	Activities	Exit output
A. Foundation	Review SRS/SDD/API/DB baselines; resolve testability; create traceability; establish CI and fixtures.	Approved plan, reconciled IDs, automated smoke gate
B. Module verification	Unit, integration, contract, and module-level system tests as features complete.	Module readiness report; defect trend
C. Cross-module system	Full lifecycle, role boundaries, concurrency, reports, notifications, archival.	System test summary
D. Non-functional	Security, performance, accessibility, resilience, backup/restore.	Specialist reports and remediation evidence
E. UAT and release	Role journeys, training check, regression, go/no-go.	Signed UAT and release decision

Cadence

Daily execution note during active testing; twice-weekly defect triage; weekly quality dashboard; immediate escalation for security, data-loss, grade-integrity, or unauthorized-disclosure defects.

Minimum status metrics

Planned/executed/pass/fail/blocked by priority and module; requirement coverage; open/ageing defects; automation stability; escaped defect rate; performance trend; environment downtime; unresolved policy decisions.

APPENDIX B

Risks and open issues

B

Risk	Impact	Mitigation / owner
Institutional policy values not ratified	Expected results change late	FYP Coordinator approves effective-dated configuration before UAT
Companion baselines renumber requirements	Traceability drift	QA Lead reconciles FR IDs at release entry
Production-like performance environment unavailable	Capacity remains uncertain	Operations reserves representative environment and documents scaling differences
Third-party notification/plagiarism sandbox instability	False failures or untested timeout paths	Contract stubs plus one controlled provider verification
Synthetic data misses real diversity	Import, accessibility, or reporting defects escape	Include Unicode, long names, nulls, boundary groups, multi-role and historical fixtures

Open decisions before formal approval

1. Approve group size limits by program/project type and whether solo projects can ever be enabled.
2. Approve supervision cap, co-supervision counting, and override authority.
3. Approve evaluation stages, weightages, rounding, moderation, appeal, and result-publication rules.
4. Approve lockout, session timeout, retention, backup RPO/RTO, and report-export limits.
5. Approve supported browsers/devices, WCAG conformance scope, and performance workload model.
6. Confirm exact SRS functional requirement IDs and API error codes beyond WF_001.

> **Baseline status.** This plan is executable with configurable fixtures. Formal production approval requires closure or explicit risk acceptance of every open decision above.